

RESEARCH PAPER

Customer Engagement & Product Utilization Analytics for Retention Strategy

Behavioral Analytics • Product Depth • Customer Engagement • Churn Risk

Prepared by

Manya

Program

B.Tech Information Technology

Internship

Unified Mentor – Data Analytics Internship
Interactive Streamlit Analytics Project

Abstract

This research paper presents a behavioral analytics framework for understanding customer retention in the banking sector. The study focuses on the premise that financial strength alone does not fully explain loyalty: customers may maintain substantial balances yet become inactive, use only a limited number of products, or show other engagement patterns associated with churn. Using a cleaned dataset of 10,000 European banking customers and 14 variables, the project evaluates engagement, product utilization, financial commitment, credit-card ownership, and churn behavior through an interactive Streamlit dashboard.

The analysis is structured around five required measures: **Engagement Retention Ratio**, **Product Depth Index**, **High-Balance Disengagement Rate**, **Credit Card Stickiness Score**, and **Relationship Strength Index**. The dashboard additionally provides core customer KPIs, interactive filters, risk segmentation, customer-level exploration, and product-utilization exploration. The project therefore creates a practical behavioral view for retention monitoring, product-bundling analysis, and identification of high-value but disengaged customer segments.

Keywords: Customer Retention, Customer Engagement, Product Utilization, Churn Analytics, Banking Analytics, Behavioral Analytics, Risk Segmentation, Streamlit, Python, Pandas, Plotly

1. Introduction

Customer retention is an important analytical concern for banks because the loss of an existing customer can affect an established relationship, product usage, and future business opportunities. Traditional analysis often emphasizes demographic or financial characteristics. However, a customer can remain financially valuable while becoming less engaged with the institution. This creates a need for a behavioral perspective that examines customer activity and relationship depth alongside financial attributes.

This project investigates the relationship between engagement and retention by examining customer activity status, number of products used, account balance, credit-card ownership, salary, and churn status. The resulting dashboard is designed as an exploratory decision-support tool that allows users to filter populations, inspect patterns, identify potentially vulnerable segments, and examine individual customer profiles.

2. Background and Business Context

The banking retention problem addressed in this study is that customers may be financially strong but still churn because of low engagement, limited product adoption, or weak relationship depth. A retention strategy therefore benefits from measuring several dimensions of the customer relationship together.

The project is organized around three central business questions:

- **Does engagement relate to retention?** Compare retention behavior between active and inactive customers.
- **Does product depth relate to loyalty?** Examine churn and retention across customers using different numbers of products.
- **Are high-value customers always engaged?** Identify customers with high balances but inactive behavior who may require retention-focused monitoring.

3. Problem Statement

Banks may lack quantitative visibility into the behavioral factors associated with customer retention. In particular, it may be difficult to determine whether deeper product relationships are associated with different churn patterns, whether high balances necessarily indicate loyalty, and which customers combine financial value with weak engagement. This project addresses these questions through structured exploratory analysis and interactive visualization.

4. Research Objectives

- Evaluate the relationship between customer engagement and churn.
- Measure the retention impact associated with product count and product mix.
- Identify inactive customers with high account balances.
- Compare retention behavior for customers with and without credit cards.
- Construct a relationship-strength perspective using engagement and product utilization.
- Create interactive customer segments for retention-focused exploration.
- Provide a practical dashboard that can support data-driven retention and product decisions.

5. Dataset and Data Description

The analysis uses the supplied European banking customer dataset. The cleaned dataset contains **10,000 records and 14 columns**. Data validation found **0 missing values, 0 duplicate rows, and 0 duplicate Customer IDs**. The Year field contains 2025 in the supplied dataset.

Field	Description
CustomerId	Unique customer identifier
Surname	Customer surname
CreditScore	Customer credit score
Geography	Customer country/region
Gender	Customer gender
Age	Customer age
Tenure	Years with the bank
Balance	Customer account balance
NumOfProducts	Number of banking products used
HasCrCard	Credit card ownership indicator
IsActiveMember	Customer activity indicator
EstimatedSalary	Estimated customer salary
Exited	Customer churn indicator
Year	Dataset year; supplied data is 2025

6. Data Validation and Preparation

Data preparation focused on validating the supplied dataset while preserving the underlying customer records. The dataset was checked for structure, missing values, duplicate rows, duplicate customer identifiers, and key-value ranges. No customer rows were removed based on the validation checks used for this project. The cleaned CSV is the analytical source for the Streamlit application.

Quality checks included:

- 10,000 customer records and 14 fields.
- 0 missing values.
- 0 duplicate rows.
- 0 duplicate Customer IDs.
- Year coverage is 2025 only.
- NumOfProducts ranges from 1 to 4.

- HasCrCard, IsActiveMember, and Exited are binary indicators.

7. Methodology

The methodology follows a structured behavioral-analytics workflow:

- **Data ingestion and validation:** Load the cleaned CSV, inspect schema, and validate data quality.
- **Engagement classification:** Separate active and inactive customers and examine combinations of activity and product depth.
- **Product utilization analysis:** Compare churn behavior across product counts and examine product depth as a relationship indicator.
- **Financial commitment versus engagement:** Analyze balance and activity together, with specific attention to high-balance inactive customers.
- **Retention strength analysis:** Combine engagement and product utilization into interpretable retention-oriented measures.
- **Interactive exploration:** Provide filters, customer-level exploration, risk segmentation, and product-specific views through Streamlit.

8. Analytical Framework and KPI Definitions

The project translates the objectives into measurable indicators. These indicators connect the dashboard's visual patterns with the research questions.

KPI	Definition / Purpose
Engagement Retention Ratio	Active-customer retention rate divided by inactive-customer retention rate; compares retention strength across engagement states.
Product Depth Index	Retained customers' average number of products relative to the overall average product count, expressed as a percentage.
High-Balance Disengagement Rate	Share of customers with balance at or above €100,000 who are inactive; used in the dashboard's risk exploration.
Credit Card Stickiness Score	Retention rate among credit-card holders minus retention rate among non-holders, expressed in percentage points.
Relationship Strength Index	Combined engagement-and-product score expressed on a 0–100 scale to summarize relationship strength.

9. Dashboard Architecture

The Streamlit application is organized as an interactive analytical interface rather than a static report. It includes a KPI layer, analytical visualizations, optional filters, and deeper exploration modules.

- **Core KPI layer:** total customers, churned customers, churn rate, active customer percentage, average products, and average balance.
- **Engagement and churn overview:** high-level retained-versus-churned and engagement analysis.
- **Product utilization analysis:** evaluates customer behavior across product-count segments.
- **High-value disengagement analysis:** examines balance and engagement together.
- **Risk segmentation:** groups filtered customers into low-, medium-, and high-risk categories using project-defined signals.
- **Customer Explorer:** enables individual Customer ID inspection.
- **Product Utilization Explorer:** allows a product-count segment to be inspected through customer count, churn rate, active rate, and average balance.

10. Interactive Filtering and User Experience

The dashboard provides optional filters for Geography, Gender, Number of Products, Balance Range, Salary Range, Engagement Status, and Customer Status. When no categorical restriction is selected, the dashboard retains the available population; when filters are applied, analytical views update to the selected population. This supports exploratory questions such as how behavior changes within a particular geography, product range, balance range, engagement state, or churn status.

11. Customer Risk Segmentation

The dashboard contains an exploratory risk score designed for the project interface. It assigns weighted signals to inactivity, low product depth, high balance, and churn. The resulting score is mapped into Low Risk, Medium Risk, and High Risk categories.

This risk segmentation is an analytical screening mechanism rather than a validated predictive model. It helps locate combinations of behavioral and financial signals that warrant closer examination within the supplied dataset.

12. Key Analytical Interpretation

The central analytical insight of the project is that retention should be examined through a combination of engagement, product utilization, and financial commitment rather than through any single variable. The dashboard is designed to reveal differences between active and inactive customers, changes in churn across product-depth segments, and cases where high balance coexists with weak engagement.

- **Engagement is a retention lens:** active and inactive customers should be compared rather than assuming financial value automatically implies loyalty.
- **Product depth provides relationship context:** number of products is used to examine whether deeper relationships coincide with different retention patterns.
- **High balance does not equal high engagement:** a high-balance inactive segment deserves separate monitoring because financial value and behavioral engagement can diverge.
- **Credit-card ownership can be evaluated as a relationship indicator:** the dashboard measures retention differences associated with card ownership without treating the relationship as causal.
- **Segment-level analysis is important:** overall churn can hide meaningful differences within product, geography, activity, and balance segments.

13. Business Implications

- **Engagement-driven retention:** monitor inactive customers and investigate whether targeted engagement programs are appropriate.
- **Product bundling decisions:** examine product-depth segments before designing cross-sell or bundling strategies.
- **Premium-customer monitoring:** track high-balance customers whose activity signals weak engagement.
- **Relationship management:** use relationship-strength indicators to prioritize analytical review of customer segments.
- **Continuous monitoring:** use interactive filters to inspect customer groups instead of relying only on an overall churn percentage.

14. Research Limitations

- The supplied dataset represents a single year (2025), so the project does not establish multi-year retention trends.

- The analysis is observational and describes associations in the supplied data; it does not establish that engagement, product count, or card ownership causes retention.
- The dashboard's risk score is a project-defined screening measure and is not a trained or validated machine-learning probability of churn.
- The dataset does not represent external factors such as service interactions, complaints, digital usage, or customer sentiment unless they are present in the supplied data.
- The €100,000 high-balance threshold is a dashboard analysis threshold and should be interpreted within the context of this dataset rather than as a universal banking standard.

15. Ethical and Responsible Use

Customer analytics should support fair, transparent, and evidence-based decision-making. Segment-level patterns should not automatically be converted into individual decisions without validation. Any operational use of retention scores or customer-risk categories should include appropriate governance, privacy controls, bias assessment, and human review. This project is an educational internship deliverable and is not intended to make automated banking decisions.

16. Technical Implementation

The dashboard is implemented in Python using Pandas for data handling, Plotly for interactive visualizations, and Streamlit for the web application interface. The application reads the cleaned CSV, creates analytical fields, calculates project KPIs, applies user-selected filters, and renders interactive metrics and charts.

Technology	Role
Python	Core programming and analytical workflow
Pandas	Data loading, cleaning, transformation and aggregation
Plotly	Interactive charts and visual analytics
Streamlit	Interactive dashboard and user interface
GitHub	Project repository and portfolio presentation

17. Conclusion

This project demonstrates a behavioral approach to banking customer-retention analytics. By combining engagement status, product utilization, balance, credit-card ownership, and churn, the analysis moves beyond a simple demographic view and provides a structured way to investigate relationship strength and potential retention risk.

The Streamlit dashboard converts the framework into an interactive tool. Users can move from high-level KPIs to filtered customer segments, product-depth comparisons, risk categories, and individual customer profiles. The framework is especially useful for identifying situations in which financial value and engagement do not align, because such cases can be difficult to recognize through balance or demographic analysis alone.

Overall, the project provides a practical foundation for retention-oriented analysis while distinguishing descriptive patterns from causal conclusions. Future extensions such as multi-year behavioral data, transaction-level activity, predictive modeling, and validated churn-probability models could strengthen the solution.

18. Deliverables and Project Outputs

- **Interactive Streamlit Dashboard:** customer engagement, product utilization, churn and risk exploration.
- **Cleaned Analytical Dataset:** European_Bank_Cleaned.csv.

- **Research Paper:** documentation of the problem, methodology, analytical framework, insights, limitations, and conclusion.
- **GitHub Repository:** source code, dataset, requirements, and documentation.
- **Executive Summary:** concise stakeholder-oriented presentation of the project and its implications.

19. Project Basis and References

The analytical framework, objectives, required KPIs, methodology, dataset fields, and deliverable structure are based on the Unified Mentor project brief and the supplied European banking customer dataset. The technical implementation uses Python, Pandas, Plotly, Streamlit, and GitHub.

This paper intentionally avoids inventing numerical findings that are not directly calculated in the dashboard. Dynamic KPI values depend on the selected filters and are therefore calculated by the application from the supplied dataset.

20. Author Information

Item	Details
Author	Manya
Program	B.Tech Information Technology
Internship	Unified Mentor – Data Analytics Internship
Project	Customer Engagement & Product Utilization Analytics for Retention Strategy
Application	Interactive Streamlit Dashboard

End of Research Paper